In [1]:
```
!pip install requests
!pip install pandas
!pip install numpy
```

```
Requirement already satisfied: requests in c:\users\michael\anaconda3a\lib\site-
packages (2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\michael\anac
onda3a\lib\site-packages (from requests) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\users\michael\anaconda3a\lib\s
ite-packages (from requests) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\michael\anaconda3a
\lib\site-packages (from requests) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\michael\anaconda3a
\lib\site-packages (from requests) (2025.1.31)
Requirement already satisfied: pandas in c:\users\michael\anaconda3a\lib\site-pa
ckages (2.2.3)
Requirement already satisfied: numpy>=1.26.0 in c:\users\michael\anaconda3a\lib
\site-packages (from pandas) (1.26.4)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\michael\anacon
da3a\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in c:\users\michael\anaconda3a\lib\s
ite-packages (from pandas) (2024.1)
Requirement already satisfied: tzdata>=2022.7 in c:\users\michael\anaconda3a\lib
\site-packages (from pandas) (2023.3)
Requirement already satisfied: six>=1.5 in c:\users\michael\anaconda3a\lib\site-
packages (from python-dateutil>=2.8.2->pandas) (1.16.0)
Requirement already satisfied: numpy in c:\users\michael\anaconda3a\lib\site-pac
kages (1.26.4)
```

In [2]:
```
# Requests allows us to make HTTP requests which we will use to get data from ar
import requests
# Pandas is a software library written for the Python programming language for d
import pandas as pd
# NumPy is a library for the Python programming language, adding support for lar
import numpy as np
# Datetime is a library that allows us to represent dates
import datetime
```

In [3]:
```
# Setting this option will print all collumns of a dataframe
pd.set_option('display.max_columns', None)
# Setting this option will print all of the data in a feature
pd.set_option('display.max_colwidth', None)
```

In [4]:
```
# Takes the dataset and uses the rocket column to call the API and append the da
def getBoosterVersion(data):
    for x in data['rocket']:
       if x:
        response = requests.get("https://api.spacexdata.com/v4/rockets/"+str(x))
        BoosterVersion.append(response['name'])
```

In [5]:
```
# Takes the dataset and uses the launchpad column to call the API and append the
def getLaunchSite(data):
    for x in data['launchpad']:
       if x:
```

```
        response = requests.get("https://api.spacexdata.com/v4/launchpads/"+st
        Longitude.append(response['longitude'])
        Latitude.append(response['latitude'])
        LaunchSite.append(response['name'])
```

In [6]:
```
# Takes the dataset and uses the payloads column to call the API and append the
def getPayloadData(data):
    for load in data['payloads']:
       if load:
        response = requests.get("https://api.spacexdata.com/v4/payloads/"+load)
        PayloadMass.append(response['mass_kg'])
        Orbit.append(response['orbit'])
```

In [7]:
```
# Takes the dataset and uses the cores column to call the API and append the dat
def getCoreData(data):
    for core in data['cores']:
            if core['core'] != None:
                response = requests.get("https://api.spacexdata.com/v4/cores/"+
                Block.append(response['block'])
                ReusedCount.append(response['reuse_count'])
                Serial.append(response['serial'])
            else:
                Block.append(None)
                ReusedCount.append(None)
                Serial.append(None)
            Outcome.append(str(core['landing_success'])+' '+str(core['landing_ty
            Flights.append(core['flight'])
            GridFins.append(core['gridfins'])
            Reused.append(core['reused'])
            Legs.append(core['legs'])
            LandingPad.append(core['landpad'])
```

In [8]:
```
spacex_url="https://api.spacexdata.com/v4/launches/past"
```

In [9]:
```
response = requests.get(spacex_url)
```

In [10]:
```
static_json_url='https://cf-courses-data.s3.us.cloud-object-storage.appdomain.cl
```

In [11]:
```
response=requests.get(static_json_url)
```

In [12]:
```
response.status_code
```

Out[12]: 200

In [13]:
```
import requests
import pandas as pd

# URL of the static JSON file
static_json_url = 'https://cf-courses-data.s3.us.cloud-object-storage.appdomain.

# Step 1: Fetch the static JSON file from the URL
response = requests.get(static_json_url)

# Step 2: Check if the request was successful
```

```
if response.status_code == 200:
    print("✅ Data fetched successfully!")
    data_json = response.json()  # Convert the response to JSON
else:
    print(f"❌ Failed to fetch data. Status code: {response.status_code}")

# Step 3: Normalize the JSON and convert to a DataFrame
data = pd.json_normalize(data_json)

# Step 4: Display DataFrame info
print("\n📊 Data shape:", data.shape)  # Displays the number of rows and column
print("\n📋 Columns:")
print(data.columns.tolist())  # Lists all the columns

# Step 5: Display first few rows
data.head()  # Shows a preview of the first few rows
```

```
✅ Data fetched successfully!

📊 Data shape: (107, 42)

📋 Columns:
['static_fire_date_utc', 'static_fire_date_unix', 'tbd', 'net', 'window', 'rocke
t', 'success', 'details', 'crew', 'ships', 'capsules', 'payloads', 'launchpad',
'auto_update', 'failures', 'flight_number', 'name', 'date_utc', 'date_unix', 'da
te_local', 'date_precision', 'upcoming', 'cores', 'id', 'fairings.reused', 'fair
ings.recovery_attempt', 'fairings.recovered', 'fairings.ships', 'links.patch.sma
ll', 'links.patch.large', 'links.reddit.campaign', 'links.reddit.launch', 'links
.reddit.media', 'links.reddit.recovery', 'links.flickr.small', 'links.flickr.ori
ginal', 'links.presskit', 'links.webcast', 'links.youtube_id', 'links.article',
'links.wikipedia', 'fairings']
```

Out[13]:

	static_fire_date_utc	static_fire_date_unix	tbd	net	window	
0	2006-03-17T00:00:00.000Z	1.142554e+09	False	False	0.0	5e9d0d95eda699
1	None	NaN	False	False	0.0	5e9d0d95eda699
2	None	NaN	False	False	0.0	5e9d0d95eda699
3	2008-09-20T00:00:00.000Z	1.221869e+09	False	False	0.0	5e9d0d95eda699

	static_fire_date_utc	static_fire_date_unix	tbd	net	window	
4	None	NaN	False	False	0.0	5e9d0d95eda699

In [23]:
```
# Lets take a subset of our dataframe keeping only the features we want and the
data = data[['rocket', 'payloads', 'launchpad', 'cores', 'flight_number', 'date_

# We will remove rows with multiple cores because those are falcon rockets with
data = data[data['cores'].map(len)==1]
data = data[data['payloads'].map(len)==1]

# Since payloads and cores are lists of size 1 we will also extract the single v
data['cores'] = data['cores'].map(lambda x : x[0])
data['payloads'] = data['payloads'].map(lambda x : x[0])

# We also want to convert the date_utc to a datetime datatype and then extracti
data['date'] = pd.to_datetime(data['date_utc']).dt.date

# Using the date we will restrict the dates of the launches
data = data[data['date'] <= datetime.date(2020, 11, 13)]
```

In [15]:
```
# Lets take a subset of our dataframe keeping only the features we want and the
data = data[['rocket', 'payloads', 'launchpad', 'cores', 'flight_number', 'date_

# We will remove rows with multiple cores because those are falcon rockets with
data = data[data['cores'].map(len)==1]
data = data[data['payloads'].map(len)==1]

# Since payloads and cores are lists of size 1 we will also extract the single v
data['cores'] = data['cores'].map(lambda x : x[0])
data['payloads'] = data['payloads'].map(lambda x : x[0])

# We also want to convert the date_utc to a datetime datatype and then extracti
data['date'] = pd.to_datetime(data['date_utc']).dt.date

# Using the date we will restrict the dates of the launches
data = data[data['date'] <= datetime.date(2020, 11, 13)]
```

In [24]:
```
#Global variables
BoosterVersion = []
```

```
PayloadMass = []
Orbit = []
LaunchSite = []
Outcome = []
Flights = []
GridFins = []
Reused = []
Legs = []
LandingPad = []
Block = []
ReusedCount = []
Serial = []
Longitude = []
Latitude = []
```

In [25]:
```
BoosterVersion
```

Out[25]: []

In [26]:
```
# Call getBoosterVersion
getBoosterVersion(data)
```

In [27]:
```
BoosterVersion[0:5]
```

Out[27]: []

In [28]:
```
# Call getLaunchSite
getLaunchSite(data)
```

In [29]:
```
# Call getPayloadData
getPayloadData(data)
```

In [30]:
```
# Call getCoreData
getCoreData(data)
```

In [32]:
```
launch_dict = {'FlightNumber': list(data['flight_number']),
'Date': list(data['date']),
'BoosterVersion':BoosterVersion,
'PayloadMass':PayloadMass,
'Orbit':Orbit,
'LaunchSite':LaunchSite,
'Outcome':Outcome,
'Flights':Flights,
'GridFins':GridFins,
'Reused':Reused,
'Legs':Legs,
'LandingPad':LandingPad,
'Block':Block,
'ReusedCount':ReusedCount,
'Serial':Serial,
'Longitude': Longitude,
'Latitude': Latitude}
```

In [33]:
```
!pip3 install beautifulsoup4
```

```
!pip3 install requests
```

```
Requirement already satisfied: beautifulsoup4 in c:\users\michael\anaconda3a\lib
\site-packages (4.12.3)
Requirement already satisfied: soupsieve>1.2 in c:\users\michael\anaconda3a\lib
\site-packages (from beautifulsoup4) (2.5)
Requirement already satisfied: requests in c:\users\michael\anaconda3a\lib\site-
packages (2.32.3)
Requirement already satisfied: charset-normalizer<4,>=2 in c:\users\michael\anac
onda3a\lib\site-packages (from requests) (3.3.2)
Requirement already satisfied: idna<4,>=2.5 in c:\users\michael\anaconda3a\lib\s
ite-packages (from requests) (3.7)
Requirement already satisfied: urllib3<3,>=1.21.1 in c:\users\michael\anaconda3a
\lib\site-packages (from requests) (2.2.3)
Requirement already satisfied: certifi>=2017.4.17 in c:\users\michael\anaconda3a
\lib\site-packages (from requests) (2025.1.31)
```

In [34]:
```
import sys

import requests
from bs4 import BeautifulSoup
import re
import unicodedata
import pandas as pd
```

In [35]:
```
def date_time(table_cells):
    """
    This function returns the data and time from the HTML  table cell
    Input: the  element of a table data cell extracts extra row
    """
    return [data_time.strip() for data_time in list(table_cells.strings)][0:2]

def booster_version(table_cells):
    """
    This function returns the booster version from the HTML  table cell
    Input: the  element of a table data cell extracts extra row
    """
    out=''.join([booster_version for i,booster_version in enumerate( table_cells
    return out

def landing_status(table_cells):
    """
    This function returns the landing status from the HTML table cell
    Input: the  element of a table data cell extracts extra row
    """
    out=[i for i in table_cells.strings][0]
    return out


def get_mass(table_cells):
    mass=unicodedata.normalize("NFKD", table_cells.text).strip()
    if mass:
        mass.find("kg")
        new_mass=mass[0:mass.find("kg")+2]
    else:
        new_mass=0
```

```
    return new_mass


def extract_column_from_header(row):
    """
    This function returns the landing status from the HTML table cell
    Input: the  element of a table data cell extracts extra row
    """
    if (row.br):
        row.br.extract()
    if row.a:
        row.a.extract()
    if row.sup:
        row.sup.extract()

    colunm_name = ' '.join(row.contents)

    # Filter the digit and empty names
    if not(colunm_name.strip().isdigit()):
        colunm_name = colunm_name.strip()
        return colunm_name
```

In [36]:
```
static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Fa
```

In [37]:
```
import requests

# The static URL of the Falcon 9 and Falcon Heavy launches Wikipedia page
static_url = "https://en.wikipedia.org/w/index.php?title=List_of_Falcon_9_and_Fa

# Perform GET request
response = requests.get(static_url)

# Check the status
print("Status Code:", response.status_code)
```

```
Status Code: 200
```

In [38]:
```
from bs4 import BeautifulSoup

# Create a BeautifulSoup object from the response content
soup = BeautifulSoup(response.text, "html.parser")

# Optional: Check the first 500 characters of the parsed HTML to verify it's wor
print(soup.prettify()[:500])  # Prints the first 500 characters in a nicely form
```

```
<!DOCTYPE html>
<html class="client-nojs vector-feature-language-in-header-enabled vector-featur
e-language-in-main-page-header-disabled vector-feature-page-tools-pinned-disable
d vector-feature-toc-pinned-clientpref-1 vector-feature-main-menu-pinned-disable
d vector-feature-limited-width-clientpref-1 vector-feature-limited-width-conten
t-enabled vector-feature-custom-font-size-clientpref-1 vector-feature-appearanc
e-pinned-clientpref-1 vector-feature-night-mode-enabled skin-theme-clientpref-da
y vect
```

In [39]:
```
# Print the title of the page
print("Page Title:", soup.title.string)
```

```
Page Title: List of Falcon 9 and Falcon Heavy launches - Wikipedia
```

In [40]:
```
# Use find_all to find all tables on the page
html_tables = soup.find_all("table")

# Check how many tables were found
print(f"Number of tables found: {len(html_tables)}")
```

```
Number of tables found: 25
```

In [41]:
```
# Let's print the third table and check its content
first_launch_table = html_tables[2]
print(first_launch_table)
```

```
<table class="wikitable plainrowheaders collapsible" style="width: 100%;">
<tbody><tr>
<th scope="col">Flight No.
</th>
<th scope="col">Date and<br/>time (<a href="/wiki/Coordinated_Universal_Time" ti
tle="Coordinated Universal Time">UTC</a>)
</th>
<th scope="col"><a href="/wiki/List_of_Falcon_9_first-stage_boosters" title="Lis
t of Falcon 9 first-stage boosters">Version,<br/>Booster</a> <sup class="referen
ce" id="cite_ref-booster_11-0"><a href="#cite_note-booster-11"><span class="cit
e-bracket">[</span>b<span class="cite-bracket">]</span></a></sup>
</th>
<th scope="col">Launch site
</th>
<th scope="col">Payload<sup class="reference" id="cite_ref-Dragon_12-0"><a href
="#cite_note-Dragon-12"><span class="cite-bracket">[</span>c<span class="cite-br
acket">]</span></a></sup>
</th>
<th scope="col">Payload mass
</th>
<th scope="col">Orbit
</th>
<th scope="col">Customer
</th>
<th scope="col">Launch<br/>outcome
</th>
<th scope="col"><a href="/wiki/Falcon_9_first-stage_landing_tests" title="Falcon
9 first-stage landing tests">Booster<br/>landing</a>
</th></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">1
</th>
<td>4 June 2010,<br/>18:45
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="r
eference" id="cite_ref-MuskMay2012_13-0"><a href="#cite_note-MuskMay2012-13"><sp
an class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><b
r/>B0003.1<sup class="reference" id="cite_ref-block_numbers_14-0"><a href="#cite
_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-br
acket">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Spa
ce Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Comp
lex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/Dragon_Spacecraft_Qualification_Unit" title="Dragon Spacecraf
t Qualification Unit">Dragon Spacecraft Qualification Unit</a>
</td>
<td>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a>
</td>
<td><a href="/wiki/SpaceX" title="SpaceX">SpaceX</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success
```

```
</td>
<td class="table-failure" style="background: #FFC7C7; color:black; vertical-alig
n: middle; text-align: center;">Failure<sup class="reference" id="cite_ref-ns201
10930_15-0"><a href="#cite_note-ns20110930-15"><span class="cite-bracket">[</spa
n>9<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_
ref-16"><a href="#cite_note-16"><span class="cite-bracket">[</span>10<span class
="cite-bracket">]</span></a></sup><br/><small>(parachute)</small>
</td></tr>
<tr>
<td colspan="9">First flight of Falcon 9 v1.0.<sup class="reference" id="cite_re
f-sfn20100604_17-0"><a href="#cite_note-sfn20100604-17"><span class="cite-bracke
t">[</span>11<span class="cite-bracket">]</span></a></sup> Used a boilerplate ve
rsion of Dragon capsule which was not designed to separate from the second stag
e.<small>(<a href="#First_flight_of_Falcon_9">more details below</a>)</small> At
tempted to recover the first stage by parachuting it into the ocean, but it burn
ed up on reentry, before the parachutes even deployed.<sup class="reference" id
="cite_ref-parachute_18-0"><a href="#cite_note-parachute-18"><span class="cite-b
racket">[</span>12<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">2
</th>
<td>8 December 2010,<br/>15:43<sup class="reference" id="cite_ref-spaceflightnow
_Clark_Launch_Report_19-0"><a href="#cite_note-spaceflightnow_Clark_Launch_Repor
t-19"><span class="cite-bracket">[</span>13<span class="cite-bracket">]</span></
a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="r
eference" id="cite_ref-MuskMay2012_13-1"><a href="#cite_note-MuskMay2012-13"><sp
an class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><b
r/>B0004.1<sup class="reference" id="cite_ref-block_numbers_14-1"><a href="#cite
_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-br
acket">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Spa
ce Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Comp
lex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_Dragon" title="SpaceX Dragon">Dragon</a> <a class="mw-
redirect" href="/wiki/COTS_Demo_Flight_1" title="COTS Demo Flight 1">demo flight
C1</a><br/>(Dragon C101)
</td>
<td>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a href="/w
iki/International_Space_Station" title="International Space Station">ISS</a>)
</td>
<td><style data-mw-deduplicate="TemplateStyles:r1126788409">.mw-parser-output .p
lainlist ol,.mw-parser-output .plainlist ul{line-height:inherit;list-style:non
e;margin:0;padding:0}.mw-parser-output .plainlist ol li,.mw-parser-output .plain
list ul li{margin-bottom:0}</style><div class="plainlist">
<ul><li><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Or
bital_Transportation_Services" title="Commercial Orbital Transportation Service
s">COTS</a>)</li>
<li><a href="/wiki/National_Reconnaissance_Office" title="National Reconnaissanc
e Office">NRO</a></li></ul>
```

```
</div>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success<sup class="reference" id="cite_ref-ns201
10930_15-1"><a href="#cite_note-ns20110930-15"><span class="cite-bracket">[</spa
n>9<span class="cite-bracket">]</span></a></sup>
</td>
<td class="table-failure" style="background: #FFC7C7; color:black; vertical-alig
n: middle; text-align: center;">Failure<sup class="reference" id="cite_ref-ns201
10930_15-2"><a href="#cite_note-ns20110930-15"><span class="cite-bracket">[</spa
n>9<span class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_
ref-20"><a href="#cite_note-20"><span class="cite-bracket">[</span>14<span class
="cite-bracket">]</span></a></sup><br/><small>(parachute)</small>
</td></tr>
<tr>
<td colspan="9">Maiden flight of <a class="mw-redirect" href="/wiki/Dragon_capsu
le" title="Dragon capsule">Dragon capsule</a>, consisting of over 3 hours of tes
ting thruster maneuvering and reentry.<sup class="reference" id="cite_ref-spacef
lightnow_Clark_unleashing_Dragon_21-0"><a href="#cite_note-spaceflightnow_Clark_
unleashing_Dragon-21"><span class="cite-bracket">[</span>15<span class="cite-bra
cket">]</span></a></sup> Attempted to recover the first stage by parachuting it
into the ocean, but it disintegrated upon reentry, before the parachutes were de
ployed.<sup class="reference" id="cite_ref-parachute_18-1"><a href="#cite_note-p
arachute-18"><span class="cite-bracket">[</span>12<span class="cite-bracket">]</
span></a></sup> <small>(<a href="#COTS_demo_missions">more details below</a>)</s
mall> It also included two <a href="/wiki/CubeSat" title="CubeSat">CubeSats</a
>,<sup class="reference" id="cite_ref-NRO_Taps_Boeing_for_Next_Batch_of_CubeSats
_22-0"><a href="#cite_note-NRO_Taps_Boeing_for_Next_Batch_of_CubeSats-22"><span
class="cite-bracket">[</span>16<span class="cite-bracket">]</span></a></sup> and
a wheel of <a href="/wiki/Brou%C3%A8re" title="Brouère">Brouère</a> cheese.
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">3
</th>
<td>22 May 2012,<br/>07:44<sup class="reference" id="cite_ref-BBC_new_era_23-0">
<a href="#cite_note-BBC_new_era-23"><span class="cite-bracket">[</span>17<span c
lass="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="r
eference" id="cite_ref-MuskMay2012_13-2"><a href="#cite_note-MuskMay2012-13"><sp
an class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><b
r/>B0005.1<sup class="reference" id="cite_ref-block_numbers_14-2"><a href="#cite
_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-br
acket">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Spa
ce Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Comp
lex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_Dragon" title="SpaceX Dragon">Dragon</a> <a class="mw-
redirect" href="/wiki/Dragon_C2%2B" title="Dragon C2+">demo flight C2+</a><sup c
lass="reference" id="cite_ref-C2_24-0"><a href="#cite_note-C2-24"><span class="c
ite-bracket">[</span>18<span class="cite-bracket">]</span></a></sup><br/>(Dragon
C102)
</td>
<td>525 kg (1,157 lb)<sup class="reference" id="cite_ref-25"><a href="#cite_not
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e-25"><span class="cite-bracket">[</span>19<span class="cite-bracket">]</span></
a></sup>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a href="/w
iki/International_Space_Station" title="International Space Station">ISS</a>)
</td>
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Orbita
l_Transportation_Services" title="Commercial Orbital Transportation Services">CO
TS</a>)
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success<sup class="reference" id="cite_ref-26"><
a href="#cite_note-26"><span class="cite-bracket">[</span>20<span class="cite-br
acket">]</span></a></sup>
</td>
<td class="table-noAttempt" style="background: #EEE; color:black; vertical-alig
n: middle; white-space: nowrap; text-align: center;">No attempt
</td></tr>
<tr>
<td colspan="9">Dragon spacecraft demonstrated a series of tests before it was a
llowed to approach the <a href="/wiki/International_Space_Station" title="Intern
ational Space Station">International Space Station</a>. Two days later, it becam
e the first commercial spacecraft to board the ISS.<sup class="reference" id="ci
te_ref-BBC_new_era_23-1"><a href="#cite_note-BBC_new_era-23"><span class="cite-b
racket">[</span>17<span class="cite-bracket">]</span></a></sup> <small>(<a href=
"#COTS_demo_missions">more details below</a>)</small>
</td></tr>
<tr>
<th rowspan="3" scope="row" style="text-align:center;">4
</th>
<td rowspan="2">8 October 2012,<br/>00:35<sup class="reference" id="cite_ref-SFN
_LLog_27-0"><a href="#cite_note-SFN_LLog-27"><span class="cite-bracket">[</span>
21<span class="cite-bracket">]</span></a></sup>
</td>
<td rowspan="2"><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><
sup class="reference" id="cite_ref-MuskMay2012_13-3"><a href="#cite_note-MuskMay
2012-13"><span class="cite-bracket">[</span>7<span class="cite-bracket">]</span>
</a></sup><br/>B0006.1<sup class="reference" id="cite_ref-block_numbers_14-3"><a
href="#cite_note-block_numbers-14"><span class="cite-bracket">[</span>8<span cla
ss="cite-bracket">]</span></a></sup>
</td>
<td rowspan="2"><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape C
anaveral Space Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space
_Launch_Complex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_CRS-1" title="SpaceX CRS-1">SpaceX CRS-1</a><sup class
="reference" id="cite_ref-sxManifest20120925_28-0"><a href="#cite_note-sxManifes
t20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracke
t">]</span></a></sup><br/>(Dragon C103)
</td>
<td>4,700 kg (10,400 lb)
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a href="/w
iki/International_Space_Station" title="International Space Station">ISS</a>)
</td>
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Resupp
```

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ly_Services" title="Commercial Resupply Services">CRS</a>)
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success
</td>
<td rowspan="2" style="background:#ececec; text-align:center;"><span class="nowr
ap">No attempt</span>
</td></tr>
<tr>
<td><a href="/wiki/Orbcomm_(satellite)" title="Orbcomm (satellite)">Orbcomm-OG2
</a><sup class="reference" id="cite_ref-Orbcomm_29-0"><a href="#cite_note-Orbcom
m-29"><span class="cite-bracket">[</span>23<span class="cite-bracket">]</span></
a></sup>
</td>
<td>172 kg (379 lb)<sup class="reference" id="cite_ref-gunter-og2_30-0"><a href
="#cite_note-gunter-og2-30"><span class="cite-bracket">[</span>24<span class="ci
te-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a>
</td>
<td><a href="/wiki/Orbcomm" title="Orbcomm">Orbcomm</a>
</td>
<td class="table-partial" style="background: #FFB; color:black; vertical-align:
middle; text-align: center;">Partial failure<sup class="reference" id="cite_ref-
nyt-20121030_31-0"><a href="#cite_note-nyt-20121030-31"><span class="cite-bracke
t">[</span>25<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<td colspan="9">CRS-1 was successful, but the <a href="/wiki/Secondary_payload"
title="Secondary payload">secondary payload</a> was inserted into an abnormally
low orbit and subsequently lost. This was due to one of the nine <a href="/wiki/
SpaceX_Merlin" title="SpaceX Merlin">Merlin engines</a> shutting down during the
launch, and NASA declining a second reignition, as per <a href="/wiki/Internatio
nal_Space_Station" title="International Space Station">ISS</a> visiting vehicle
safety rules, the primary payload owner is contractually allowed to decline a se
cond reignition. NASA stated that this was because SpaceX could not guarantee a
high enough likelihood of the second stage completing the second burn successful
ly which was required to avoid any risk of secondary payload's collision with th
e ISS.<sup class="reference" id="cite_ref-OrbcommTotalLoss_32-0"><a href="#cite_
note-OrbcommTotalLoss-32"><span class="cite-bracket">[</span>26<span class="cit
e-bracket">]</span></a></sup><sup class="reference" id="cite_ref-sn20121011_3
3-0"><a href="#cite_note-sn20121011-33"><span class="cite-bracket">[</span>27<sp
an class="cite-bracket">]</span></a></sup><sup class="reference" id="cite_ref-3
4"><a href="#cite_note-34"><span class="cite-bracket">[</span>28<span class="cit
e-bracket">]</span></a></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">5
</th>
<td>1 March 2013,<br/>15:10
</td>
<td><a href="/wiki/Falcon_9_v1.0" title="Falcon 9 v1.0">F9 v1.0</a><sup class="r
eference" id="cite_ref-MuskMay2012_13-4"><a href="#cite_note-MuskMay2012-13"><sp
an class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><b
r/>B0007.1<sup class="reference" id="cite_ref-block_numbers_14-4"><a href="#cite
_note-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-br
```

```
acket">]</span></a></sup>
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Spa
ce Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Comp
lex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
<td><a href="/wiki/SpaceX_CRS-2" title="SpaceX CRS-2">SpaceX CRS-2</a><sup class
="reference" id="cite_ref-sxManifest20120925_28-1"><a href="#cite_note-sxManifes
t20120925-28"><span class="cite-bracket">[</span>22<span class="cite-bracke
t">]</span></a></sup><br/>(Dragon C104)
</td>
<td>4,877 kg (10,752 lb)
</td>
<td><a href="/wiki/Low_Earth_orbit" title="Low Earth orbit">LEO</a> (<a class="m
w-redirect" href="/wiki/ISS" title="ISS">ISS</a>)
</td>
<td><a href="/wiki/NASA" title="NASA">NASA</a> (<a href="/wiki/Commercial_Resupp
ly_Services" title="Commercial Resupply Services">CRS</a>)
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success
</td>
<td class="table-noAttempt" style="background: #EEE; color:black; vertical-alig
n: middle; white-space: nowrap; text-align: center;">No attempt
</td></tr>
<tr>
<td colspan="9">Last launch of the original Falcon 9 v1.0 <a href="/wiki/Launch_
vehicle" title="Launch vehicle">launch vehicle</a>, first use of the unpressuriz
ed trunk section of Dragon.<sup class="reference" id="cite_ref-sxf9_20110321_3
5-0"><a href="#cite_note-sxf9_20110321-35"><span class="cite-bracket">[</span>29
<span class="cite-bracket">]</span></a></sup>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">6
</th>
<td>29 September 2013,<br/>16:00<sup class="reference" id="cite_ref-pa20130930_3
6-0"><a href="#cite_note-pa20130930-36"><span class="cite-bracket">[</span>30<sp
an class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a><sup class="r
eference" id="cite_ref-MuskMay2012_13-5"><a href="#cite_note-MuskMay2012-13"><sp
an class="cite-bracket">[</span>7<span class="cite-bracket">]</span></a></sup><b
r/>B1003<sup class="reference" id="cite_ref-block_numbers_14-5"><a href="#cite_n
ote-block_numbers-14"><span class="cite-bracket">[</span>8<span class="cite-brac
ket">]</span></a></sup>
</td>
<td><a class="mw-redirect" href="/wiki/Vandenberg_Air_Force_Base" title="Vandenb
erg Air Force Base">VAFB</a>,<br/><a href="/wiki/Vandenberg_Space_Launch_Complex
_4" title="Vandenberg Space Launch Complex 4">SLC-4E</a>
</td>
<td><a href="/wiki/CASSIOPE" title="CASSIOPE">CASSIOPE</a><sup class="reference"
id="cite_ref-sxManifest20120925_28-2"><a href="#cite_note-sxManifest20120925-2
8"><span class="cite-bracket">[</span>22<span class="cite-bracket">]</span></a>
</sup><sup class="reference" id="cite_ref-CASSIOPE_MDA_37-0"><a href="#cite_not
e-CASSIOPE_MDA-37"><span class="cite-bracket">[</span>31<span class="cite-bracke
t">]</span></a></sup>
```

```
</td>
<td>500 kg (1,100 lb)
</td>
<td><a href="/wiki/Polar_orbit" title="Polar orbit">Polar orbit</a> <a href="/wi
ki/Low_Earth_orbit" title="Low Earth orbit">LEO</a>
</td>
<td><a href="/wiki/Maxar_Technologies" title="Maxar Technologies">MDA</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success<sup class="reference" id="cite_ref-pa201
30930_36-1"><a href="#cite_note-pa20130930-36"><span class="cite-bracket">[</spa
n>30<span class="cite-bracket">]</span></a></sup>
</td>
<td class="table-no2" style="background: #FFE3E3; color: black; vertical-align:
middle; text-align: center;">Uncontrolled<br/><small>(ocean)</small><sup class="
reference" id="cite_ref-ocean_landing_38-0"><a href="#cite_note-ocean_landing-3
8"><span class="cite-bracket">[</span>d<span class="cite-bracket">]</span></a></
sup>
</td></tr>
<tr>
<td colspan="9">First commercial mission with a private customer, first launch f
rom Vandenberg, and demonstration flight of Falcon 9 v1.1 with an improved 13-to
nne to LEO capacity.<sup class="reference" id="cite_ref-sxf9_20110321_35-1"><a h
ref="#cite_note-sxf9_20110321-35"><span class="cite-bracket">[</span>29<span cla
ss="cite-bracket">]</span></a></sup> After separation from the second stage carr
ying Canadian commercial and scientific satellites, the first stage booster perf
ormed a controlled reentry,<sup class="reference" id="cite_ref-39"><a href="#cit
e_note-39"><span class="cite-bracket">[</span>32<span class="cite-bracket">]</sp
an></a></sup> and an <a href="/wiki/Falcon_9_first-stage_landing_tests" title="F
alcon 9 first-stage landing tests">ocean touchdown test</a> for the first time.
This provided good test data, even though the booster started rolling as it near
ed the ocean, leading to the shutdown of the central engine as the roll depleted
it of fuel, resulting in a hard impact with the ocean.<sup class="reference" id
="cite_ref-pa20130930_36-2"><a href="#cite_note-pa20130930-36"><span class="cit
e-bracket">[</span>30<span class="cite-bracket">]</span></a></sup> This was the
first known attempt of a rocket engine being lit to perform a supersonic retro p
ropulsion, and allowed SpaceX to enter a public-private partnership with <a href
="/wiki/NASA" title="NASA">NASA</a> and its Mars entry, descent, and landing tec
hnologies research projects.<sup class="reference" id="cite_ref-40"><a href="#ci
te_note-40"><span class="cite-bracket">[</span>33<span class="cite-bracket">]</s
pan></a></sup> <small>(<a href="#Maiden_flight_of_v1.1">more details below</a
>)</small>
</td></tr>
<tr>
<th rowspan="2" scope="row" style="text-align:center;">7
</th>
<td>3 December 2013,<br/>22:41<sup class="reference" id="cite_ref-sfn_wwls201306
24_41-0"><a href="#cite_note-sfn_wwls20130624-41"><span class="cite-bracket">[</
span>34<span class="cite-bracket">]</span></a></sup>
</td>
<td><a href="/wiki/Falcon_9_v1.1" title="Falcon 9 v1.1">F9 v1.1</a><br/>B1004
</td>
<td><a href="/wiki/Cape_Canaveral_Space_Force_Station" title="Cape Canaveral Spa
ce Force Station">CCAFS</a>,<br/><a href="/wiki/Cape_Canaveral_Space_Launch_Comp
lex_40" title="Cape Canaveral Space Launch Complex 40">SLC-40</a>
</td>
```

```
<td><a href="/wiki/SES-8" title="SES-8">SES-8</a><sup class="reference" id="cite
_ref-sxManifest20120925_28-3"><a href="#cite_note-sxManifest20120925-28"><span c
lass="cite-bracket">[</span>22<span class="cite-bracket">]</span></a></sup><sup
class="reference" id="cite_ref-spx-pr_42-0"><a href="#cite_note-spx-pr-42"><span
class="cite-bracket">[</span>35<span class="cite-bracket">]</span></a></sup><sup
class="reference" id="cite_ref-aw20110323_43-0"><a href="#cite_note-aw20110323-4
3"><span class="cite-bracket">[</span>36<span class="cite-bracket">]</span></a>
</sup>
</td>
<td>3,170 kg (6,990 lb)
</td>
<td><a href="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer o
rbit">GTO</a>
</td>
<td><a class="mw-redirect" href="/wiki/SES_S.A." title="SES S.A.">SES</a>
</td>
<td class="table-success" style="background: #9EFF9E; color:black; vertical-alig
n: middle; text-align: center;">Success<sup class="reference" id="cite_ref-SNMis
sionStatus7_44-0"><a href="#cite_note-SNMissionStatus7-44"><span class="cite-bra
cket">[</span>37<span class="cite-bracket">]</span></a></sup>
</td>
<td class="table-noAttempt" style="background: #EEE; color:black; vertical-alig
n: middle; white-space: nowrap; text-align: center;">No attempt<br/><sup class="
reference" id="cite_ref-sf10120131203_45-0"><a href="#cite_note-sf10120131203-4
5"><span class="cite-bracket">[</span>38<span class="cite-bracket">]</span></a>
</sup>
</td></tr>
<tr>
<td colspan="9">First <a href="/wiki/Geostationary_transfer_orbit" title="Geosta
tionary transfer orbit">Geostationary transfer orbit</a> (GTO) launch for Falcon
9,<sup class="reference" id="cite_ref-spx-pr_42-1"><a href="#cite_note-spx-pr-4
2"><span class="cite-bracket">[</span>35<span class="cite-bracket">]</span></a>
</sup> and first successful reignition of the second stage.<sup class="reference
" id="cite_ref-46"><a href="#cite_note-46"><span class="cite-bracket">[</span>39
<span class="cite-bracket">]</span></a></sup> SES-8 was inserted into a <a href
="/wiki/Geostationary_transfer_orbit" title="Geostationary transfer orbit">Supe
r-Synchronous Transfer Orbit</a> of 79,341 km (49,300 mi) in apogee with an <a h
ref="/wiki/Orbital_inclination" title="Orbital inclination">inclination</a> of 2
0.55° to the <a href="/wiki/Equator" title="Equator">equator</a>.
</td></tr></tbody></table>
```

In [42]:

```python
# Initialize an empty list to hold the column names
column_names = []

# Iterate through all <th> elements in the first_launch_table
for th in first_launch_table.find_all("th"):
    # Apply the provided extract_column_from_header() function to extract the co
    column_name = extract_column_from_header(th)

    # Append the non-empty column name to the list
    if column_name and len(column_name) > 0:
        column_names.append(column_name)

# Print the column names
print("Column Names:", column_names)
```

```
Column Names: ['Flight No.', 'Date and time ( )', 'Launch site', 'Payload', 'Pay
load mass', 'Orbit', 'Customer', 'Launch outcome']
```

In [43]:
```
print(column_names)
```

```
['Flight No.', 'Date and time ( )', 'Launch site', 'Payload', 'Payload mass', 'O
rbit', 'Customer', 'Launch outcome']
```

In [44]:
```
# Initialize an empty dictionary with column names as keys
launch_dict = dict.fromkeys(column_names)

# Remove the irrelevant column
del launch_dict['Date and time ( )']

# Initialize each column with an empty list
launch_dict['Flight No.'] = []
launch_dict['Launch site'] = []
launch_dict['Payload'] = []
launch_dict['Payload mass'] = []
launch_dict['Orbit'] = []
launch_dict['Customer'] = []
launch_dict['Launch outcome'] = []

# Add the new columns
launch_dict['Version Booster'] = []
launch_dict['Booster landing'] = []
launch_dict['Date'] = []
launch_dict['Time'] = []

# Print the dictionary to check its structure
print(launch_dict)
```

```
{'Flight No.': [], 'Launch site': [], 'Payload': [], 'Payload mass': [], 'Orbit'
: [], 'Customer': [], 'Launch outcome': [], 'Version Booster': [], 'Booster land
ing': [], 'Date': [], 'Time': []}
```

In [45]:
```
# Create the DataFrame from the dictionary
df = pd.DataFrame({key: pd.Series(value) for key, value in launch_dict.items()})

# Display the first few rows of the DataFrame to verify
print(df.head())
```

```
Empty DataFrame
Columns: [Flight No., Launch site, Payload, Payload mass, Orbit, Customer, Launc
h outcome, Version Booster, Booster landing, Date, Time]
Index: []
```

In [46]:
```
# Export the DataFrame to a CSV file
df.to_csv('spacex_web_scraped.csv', index=False)

# Confirm the CSV has been saved
print("CSV file 'spacex_web_scraped.csv' has been successfully saved.")
```

```
CSV file 'spacex_web_scraped.csv' has been successfully saved.
```

In [47]:
```
# Filter the dataframe to only include Falcon 9 launches
falcon_9_df = df[df['Payload'].str.contains('Falcon 9', na=False)]
```

```
# Display the filtered dataframe
print(falcon_9_df.head())
```

```
Empty DataFrame
Columns: [Flight No., Launch site, Payload, Payload mass, Orbit, Customer, Launc
h outcome, Version Booster, Booster landing, Date, Time]
Index: []
```

In [48]:
```
# Replace NaN values in 'Payload mass' with the mean of that column
df['Payload mass'].fillna(df['Payload mass'].mean(), inplace=True)

# Display the updated DataFrame to verify
print(df['Payload mass'].head())
```

```
Series([], Name: Payload mass, dtype: object)
C:\Users\Michael\AppData\Local\Temp\ipykernel_14288\1234062768.py:2: FutureWarni
ng: A value is trying to be set on a copy of a DataFrame or Series through chain
ed assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work beca
use the intermediate object on which we are setting values always behaves as a c
opy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.met
hod({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to
perform the operation inplace on the original object.


  df['Payload mass'].fillna(df['Payload mass'].mean(), inplace=True)
```

In [49]:
```
# Remove 'kg' and commas, then convert to numeric values
df['Payload mass'] = df['Payload mass'].str.replace(' kg', '', regex=False)  # R
df['Payload mass'] = df['Payload mass'].str.replace(',', '', regex=False)  # Re

# Convert the column to numeric values (this will convert invalid parsing to NaN
df['Payload mass'] = pd.to_numeric(df['Payload mass'], errors='coerce')

# Replace NaN values with the mean of the column
df['Payload mass'].fillna(df['Payload mass'].mean(), inplace=True)

# Display the updated DataFrame to verify
print(df['Payload mass'].head())
```

```
Series([], Name: Payload mass, dtype: int64)
C:\Users\Michael\AppData\Local\Temp\ipykernel_14288\4205711417.py:9: FutureWarni
ng: A value is trying to be set on a copy of a DataFrame or Series through chain
ed assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work beca
use the intermediate object on which we are setting values always behaves as a c
opy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.met
hod({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to
perform the operation inplace on the original object.


  df['Payload mass'].fillna(df['Payload mass'].mean(), inplace=True)
```

In [50]:
```
# Remove 'kg' and commas, then convert to numeric values
df['Payload mass'] = df['Payload mass'].str.replace(' kg', '', regex=False)  # I
df['Payload mass'] = df['Payload mass'].str.replace(',', '', regex=False)  # Re

# Convert the column to numeric values (this will convert invalid parsing to NaN
df['Payload mass'] = pd.to_numeric(df['Payload mass'], errors='coerce')

# Replace NaN values with the mean of the column
df['Payload mass'].fillna(df['Payload mass'].mean(), inplace=True)

# Display the updated DataFrame to verify
print(df['Payload mass'].head())
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[50], line 2
      1 # Remove 'kg' and commas, then convert to numeric values
----> 2 df['Payload mass'] = df['Payload mass'].str.replace(' kg', '', regex=Fal
se)  # Remove ' kg'
      3 df['Payload mass'] = df['Payload mass'].str.replace(',', '', regex=Fals
e)  # Remove commas
      5 # Convert the column to numeric values (this will convert invalid parsin
g to NaN)

File ~\anaconda3\Lib\site-packages\pandas\core\generic.py:6299, in NDFrame.__get
attr__(self, name)
   6292 if (
   6293     name not in self._internal_names_set
   6294     and name not in self._metadata
   6295     and name not in self._accessors
   6296     and self._info_axis._can_hold_identifiers_and_holds_name(name)
   6297 ):
   6298     return self[name]
-> 6299 return object.__getattribute__(self, name)

File ~\anaconda3\Lib\site-packages\pandas\core\accessor.py:224, in CachedAccesso
r.__get__(self, obj, cls)
    221 if obj is None:
    222     # we're accessing the attribute of the class, i.e., Dataset.geo
    223     return self._accessor
--> 224 accessor_obj = self._accessor(obj)
    225 # Replace the property with the accessor object. Inspired by:
    226 # https://www.pydanny.com/cached-property.html
    227 # We need to use object.__setattr__ because we overwrite __setattr__ on
    228 # NDFrame
    229 object.__setattr__(obj, self._name, accessor_obj)

File ~\anaconda3\Lib\site-packages\pandas\core\strings\accessor.py:191, in Strin
gMethods.__init__(self, data)
    188 def __init__(self, data) -> None:
    189     from pandas.core.arrays.string_ import StringDtype
--> 191     self._inferred_dtype = self._validate(data)
    192     self._is_categorical = isinstance(data.dtype, CategoricalDtype)
    193     self._is_string = isinstance(data.dtype, StringDtype)

File ~\anaconda3\Lib\site-packages\pandas\core\strings\accessor.py:245, in Strin
gMethods._validate(data)
    242 inferred_dtype = lib.infer_dtype(values, skipna=True)
    244 if inferred_dtype not in allowed_types:
--> 245     raise AttributeError("Can only use .str accessor with string value
s!")
    246 return inferred_dtype

AttributeError: Can only use .str accessor with string values!
```

In [51]:
```
# Filter the dataframe to only include Falcon 9 launches
falcon_9_df = df[df['Payload'].str.contains('Falcon 9', na=False)]

# Display the filtered dataframe
```

```
print(falcon_9_df.head())
```

```
Empty DataFrame
Columns: [Flight No., Launch site, Payload, Payload mass, Orbit, Customer, Launc
h outcome, Version Booster, Booster landing, Date, Time]
Index: []
```

In [52]:
```
# Replace NaN values in 'Payload mass' with the mean of that column
df['Payload mass'].fillna(df['Payload mass'].mean(), inplace=True)

# Display the updated DataFrame to verify
print(df['Payload mass'].head())
```

```
Series([], Name: Payload mass, dtype: int64)
C:\Users\Michael\AppData\Local\Temp\ipykernel_14288\1234062768.py:2: FutureWarni
ng: A value is trying to be set on a copy of a DataFrame or Series through chain
ed assignment using an inplace method.
The behavior will change in pandas 3.0. This inplace method will never work beca
use the intermediate object on which we are setting values always behaves as a c
opy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.met
hod({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to
perform the operation inplace on the original object.
```

```
df['Payload mass'].fillna(df['Payload mass'].mean(), inplace=True)
```

In [53]:
```
import requests
import pandas as pd

# Step 1: Send GET request to the SpaceX API
url = "https://api.spacexdata.com/v4/launches"  # SpaceX API URL for launches
response = requests.get(url)

# Step 2: Convert the response JSON into a DataFrame
launch_data = response.json()
df = pd.json_normalize(launch_data)

# Step 3: Filter for Falcon 9 launches (assuming the vehicle info is in a key co
# This assumes the column with the rocket type name is 'rocket' and Falcon 9 lau
falcon_9_df = df[df['rocket'].str.contains('Falcon 9', case=False, na=False)]

# Step 4: Remove Falcon 1 launches by filtering them out (assuming 'Falcon 1' is
falcon_9_df = falcon_9_df[~falcon_9_df['rocket'].str.contains('Falcon 1', case=F

# Step 5: Count the number of Falcon 9 launches
falcon_9_launch_count = len(falcon_9_df)

# Print the count of Falcon 9 launches after removing Falcon 1 launches
print(f"Number of Falcon 9 launches after removing Falcon 1 launches: {falcon_9_
```

```
Number of Falcon 9 launches after removing Falcon 1 launches: 0
```

In [ ]: